A Supplemental Web Appendix

The appendix includes proofs of the theorems stated in the main text, as well as statements and proofs of supplemental lemmata. All notation is as established in the main text, unless stated otherwise. Equation numbering continues in sequence with that established in the main text.

In addition to proofs, Section A.5 of the Appendix summarizes the results of a small set of Monte Carlo experiments and Section A.6 discusses additional applications of our MCMC algorithm.

A.1 Measurability of the likelihood

For the equation (9) to be well-defined we must show that $\mathcal{N}(\mathbf{d}, \cdot; \theta)$ is measurable. For any network $\mathbf{d}$ we can define a function $\mathcal{N}(\mathbf{d}, \cdot; \theta)$, which assigns to the realization $\mathbf{U} = \mathbf{u}$ a probability weight for the pure strategy combination which corresponds to $\mathbf{d}$. We now show that there is a measurable function $\mathcal{N}(\mathbf{d}, \cdot; \theta)$ satisfying these conditions.

In order to show measurability, we use random set theory. Let X be the set of NE in the game. This set depends on the random realization of the taste shock. We now use Proposition 3.1 of Beresteanu et al. (2011) to show that X is a random closed set. We verify that assumptions (i) to (iii) of their Proposition 3.1 are fulfilled.

- **(i)** The game is a network formation game, which implies the finiteness of the outcome set and that each player has only a finite number of strategies. This ensures the first assumption.

- **(ii)** The second assumption states that the observed outcome of the game results from static, simultaneous-move Nash play, which is ensured by the structure of our game's setup.

- **(iii)** The payoff functions are known for a given strategy and, without loss of generality, can be assumed to be normalized. The payoff functions are continuous in covariates and the taste shocks, which follows directly from equation (2). Without loss of generality, we can assume the parameters belong to a compact subset of the real cube, as the parameters are indicator vectors, constants, and fixed for the test.

With these assumptions, Proposition 3.1 states that X is a random closed set.

We know from Nash's existence theorem that for each game there exists a NE. Therefore X is nonempty. The set of NE is a closed set. We therefore can apply Molchanov's (2017) Fundamental Selection Theorem (Theorem 1.4.1 on p. 77) and find a measurable selection $\xi : \mathbb{R}^n \to \Sigma$ which assigns to each taste shock $\mathbf{u}$, a NE.

Let $h_{\mathbf{d}} : \Sigma \to [0,1]$ be the function which assigns to every mixed strategy the probability of $\mathbf{d}$ by multiplying the mixed strategies weights corresponding to $\mathbf{d}$. Since multiplication is a measurable operation $h_{\mathbf{d}}$ is measurable and $\mathcal{N}(\mathbf{d}, \cdot; \theta) := h_{\mathbf{d}} \circ \xi$ satisfies the desired properties.

A.2 Proof of Lemma 2.1

Let $c(\delta) = \left[\prod_{i=1}^{N} \prod_{j \neq i} \left(1 + \exp\left(W_{ij}'\lambda + R_i'\mathbf{A} + R_j'\mathbf{B}\right)\right)\right]^{-1}$; algebraic manipulation of (12) in the main text yields

$$\begin{aligned}
P_0(\mathbf{d};\delta) =& \prod_{i=1}^{N}\prod_{j\neq i}\left[\frac{\exp\left(W_{ij}'\lambda + R_i'\mathbf{A} + R_j'\mathbf{B}\right)}{1+\exp\left(W_{ij}'\lambda + R_i'\mathbf{A} + R_j'\mathbf{B}\right)}\right]^{d_{ij}} \times \left[\frac{1}{1+\exp\left(W_{ij}'\lambda + R_i'\mathbf{A} + R_j'\mathbf{B}\right)}\right]^{1-d_{ij}} \\
=& \prod_{i=1}^{N}\prod_{j\neq i}\frac{\exp\left(W_{ij}'\lambda + R_i'\mathbf{A} + R_j'\mathbf{B}\right)^{d_{ij}}}{1+\exp\left(W_{ij}'\lambda + R_i'\mathbf{A} + R_j'\mathbf{B}\right)} \\
=& c(\delta)\exp\left(\left[\sum_{i=1}^{N}\sum_{j\neq i} d_{ij}W_{ij}\right]'\lambda\right) \\
& \times \exp\left(\left[\sum_{i=1}^{N}\sum_{j\neq i} d_{ij}R_i\right]'\mathbf{A}\right)\exp\left(\left[\sum_{i=1}^{N}\sum_{j\neq i} d_{ij}R_j\right]'\mathbf{B}\right) \\
=& c(\delta)\exp\left(\mathrm{vec}\left(\mathbf{m}'\right)'\lambda\right)\exp\left(\mathbf{s}_{\text{out}}'\mathbf{A}\right)\exp\left(\mathbf{s}_{\text{in}}'\mathbf{B}\right) \\
=& c(\delta)\exp\left(\mathbf{t}'\delta\right),
\end{aligned}$$

which belongs to the exponential family of distributions with a sufficient statistic for δ of $\mathbf{T}$ (see, for example, Definition 1 in Ferguson (1967, p. 126)).

That $\mathbf{T}$ is a boundedly complete sufficient statistic for δ follows from the observation that, if $k(\mathbf{t})$ is a bounded real-valued function for which $\mathbb{E}_0[k(\mathbf{T})] = 0$ for all $\delta \in \Delta$, then

$$c(\delta)\sum_{\mathbf{t}\in\mathbb{T}} k(\mathbf{t})\exp\left(\mathbf{t}'\delta\right) \equiv 0, \text{ for all } \delta \in \Delta. \tag{35}$$

Since the range of the exponential function is the set of all positive real numbers and $c(\delta) > 0$ for all $\delta \in \Delta$, we have $\mathbb{E}_0[k(\mathbf{T})] = 0$ for all $\delta \in \Delta$ implies that $\Pr_0(k(\mathbf{T}) = 0) = 1$ (recall that Δ includes an open rectangle, hence if the summands in (35) exactly sum to zero for a particular δ in the open rectangle, they will not do so for adjacent values of δ by continuity). We have that $\mathbf{T}$ is a complete sufficient statistic for δ. Minimal sufficiency follows from completeness. See Ferguson (1967, Section 1, pp. 132 - 136) for additional background and related results.

To get the form of $c(\delta)$ given in the statement of the Lemma observe that

$$\begin{aligned}
c(\delta)^{-1} &= \left[\prod_{i=1}^{N}\prod_{j\neq i}\left(1+\exp\left(W_{ij}'\lambda+R_i'\mathbf{A}+R_j'\mathbf{B}\right)\right)\right] \\
&= \left[\prod_{i=1}^{N}\prod_{j\neq i}\left(\exp(0)+\exp\left(W_{ij}'\lambda+R_i'\mathbf{A}+R_j'\mathbf{B}\right)\right)\right] \\
&= \sum_{d\in\mathbb{D}_N}\exp\left(\left[\sum_{i=1}^{N}\sum_{j\neq i}d_{ij}\left\{W_{ij}'\lambda+R_i'\mathbf{A}+R_j'\mathbf{B}\right\}\right]\right).
\end{aligned}$$

A.3 Derivation of locally best test statistic

We begin with a high level overview of our argument; with a formal proof immediately following.

One feature of $s_{ij}(\mathbf{d})$, which will prove central to our analysis, is that is has finite range. To see this observe that since the set of all networks $\mathbb{D}_N$ is finite, $s_{ij}(\mathbf{d})$ also takes only a finite number of values. Let $\mathbb{S}=\{\underline{s},s_1,\ldots,s_M,\overline{s}\}$ be the set of possible values for $s_{ij}(\mathbf{d})$, ordered from smallest to largest.

An example illustrates. If $s_{ij}(\mathbf{d})=d_{ji}$, as occurs when agents prefer reciprocated links, we have $\mathbb{S}=\{0,1\}$. If $s_{ij}(\mathbf{d})=\sum_k d_{ki}d_{kj}$, as when agents prefer supported links, we have $\mathbb{S}=\{0,1,\ldots,N-2\}$. Finiteness of the cardinality of $\mathbb{S}$ (for a given N) plays an important role in our analysis, as will become apparent below.

To understand the likelihood (9) it is helpful to consider a (relatively) simple example. This example will also help in understanding our derivation of the optimal test statistic below. Assume that $s_{ij}(\mathbf{d})=d_{ji}$ such that agents prefer reciprocated links when $\gamma>0$. In this example $s_{ij}(\mathbf{d})$ equals either zero (j does not reciprocate) or one (j does reciprocate). We can use the two elements of $\mathbb{S}$ to partition the real line into what we will call *buckets*:

$$\mathbb{R}=(-\infty,\mu_{ij}]\cup(\mu_{ij},\mu_{ij}+\gamma]\cup(\mu_{ij}+\gamma,\infty). \tag{36}$$

Here $\mu_{ij}=A_i+B_j+X_j'\Lambda_0X_i$ equals the systematic component of baseline utility generated by arc ij. Next consider the realization of U_{ij}, the idiosyncratic utility agent i gets when she directs a link to j. If U_{ij} falls into the first bucket in (36), then agent i will always direct a link to j; irrespective of whether j chooses to direct a link to i or not. If U_{ij} falls into the middle or *inner* bucket, however, then i will direct a link to j only if j reciprocates. Finally, if U_{ij} falls into the last bucket, then i will never direct a link to j regardless of whether j directs a link to i or not. We will call the first and last buckets in (36) *outer* buckets.

If both U_{ij} and U_{ji} fall in their respective *inner* buckets, then the $\{i, j\}$ dyad can either take the empty ($D_{ij} = D_{ji} = 0$) or reciprocated ($D_{ij} = D_{ji} = 1$) configuration in equilibrium. In contrast, if *either* U_{ij} or U_{ji} falls into an outer bucket, then the $\{i, j\}$ dyad's wiring is uniquely determined. For example if U_{ij} is in the first outer bucket and U_{ji} is in the inner bucket, then the $\{i, j\}$ dyad will take the reciprocated form with probability one. It is a strictly dominant strategy for i to direct an link to j in this case and a best response for j to reciprocate.

For $\mathbf{U} = \mathbf{u}$ and $\theta = \theta_0$, let $J(\mathbf{u}; \theta_0) \leq \binom{N}{2}$ equal the number of dyads $\{i, j\}$, where both u_{ij} and u_{ji} fall into their inner bucket. For each of these dyads both the empty and reciprocated configuration is an equilibrium outcome. There are therefore $2^{J(\mathbf{u};\theta_0)}$ equilibrium networks in this case; the $\mathcal{N}(\mathbf{d}, \mathbf{u}; \theta_0)$ function would assign some probability between zero and one to each of these $2^{J(\mathbf{u};\theta_0)}$ networks (summing to one in total).

Recall that $\mathbb{S} = \{\underline{s}, s_1, \ldots, s_M, \overline{s}\}$ equals the possible values of $s_{ij}(\mathbf{d})$, arranged from smallest to largest. We can use these support points to partition $\mathbb{R}$ into a set of intervals $\mathbb{B}$:

$$\mathbb{R} = (-\infty, \mu_{ij} + \gamma\underline{s}] \cup (\mu_{ij} + \gamma\underline{s}, \mu_{ij} + \gamma s_1] \cup \cdots \cup (\mu_{ij} + \gamma s_M, \mu_{ij} + \gamma\overline{s}] \cup (\mu_{ij} + \gamma\overline{s}, \infty) \,. \quad (37)$$

The elements of $\mathbb{B}$, called *buckets*, correspond to the intervals listed in (37). In principle we should write $\mathbb{B}_{ij}$ instead of $\mathbb{B}$, reflecting the dependence of the bucket definitions on the value of μ_{ij}, the systematic non-strategic utility associated with an i-to-j link. However, since this dependence is not essential to any of the arguments that follow we leave it implicit. Note that the cardinality of $\mathbb{B}$ does not depend on μ_{ij}, but instead equals $|\mathbb{S}| + 1$.

Agent i's linking behavior vis-a-vis j depends on which bucket U_{ij} falls into. For $B \in \mathbb{B}$, if $U_{ij} \in B$, then we say U_{ij} is in, or falls into, bucket B. The first and last buckets, respectively $(-\infty, \mu_{ij} + \gamma\underline{s}]$ and $(\mu_{ij} + \gamma\overline{s}, \infty)$, play an important role in our argument. We call these two buckets *outer buckets*. The rest of the buckets we call *inner buckets*.

If U_{ij} falls into one of these outer buckets then player i has a pure strategy for d_{ij} which is strictly dominating. Specifically if U_{ij} falls into the lowest bucket, then i will direct an link to j regardless of what actions are taken by the other agents in the network. The marginal utility generated by link ij is so large that it remains positive across all possible configurations of the rest of the network; hence i always chooses to direct an link to j.

If, instead, U_{ij} falls into the highest bucket, then i will never direct an link to j. In this case the marginal utility associated with link ij is so low that it remains negative across all possible configurations of the rest of the network; hence i never chooses to direct a link to j.

Finally, if U_{ij} falls into an inner bucket, say $(\mu_{ij} + \gamma s_m, \mu_{ij} + \gamma s_{m+1}]$, then agent i's optimal choice for d_{ij} is contingent upon the linking behavior of other agents. If other agents' link actions are such that $s_{ij}(\mathbf{d}) \geq s_m$, then it is a best response for i to link with j, but not otherwise.

The vector of idiosyncratic taste shocks, $\mathbf{U}$ contains $n = N(N-1)$ elements; one for each possible arc. Let the boldface subscripts $\mathbf{i} = \mathbf{1}, \mathbf{2}, \ldots$ index these potential arcs in arbitrary order (e.g., $\mathbf{i}$ maps to some ij and vice-versa). Let $\mathbf{b} \in \mathbb{B}^n \overset{def}{\equiv} \mathbb{B} \times \cdots \times \mathbb{B}$ and $\mathbf{U} = (U_{\mathbf{1}}, \ldots, U_{\mathbf{n}})'$; we have that $\mathbf{U} \in \mathbf{b}$ for $\mathbf{b} \in \mathbb{B}^n$ so that each element of $\mathbf{u}$ falls into a bucket.

With the above notation established we can rewrite the likelihood (9) as:

$$P(\mathbf{d}; \theta, \mathcal{N}) = \sum_{\mathbf{b} \in \mathbb{B}^n} \int_{\mathbf{u} \in \mathbf{b}} \mathcal{N}(\mathbf{d}, \mathbf{u}; \theta) f_{\mathbf{U}}(\mathbf{u}) \, \mathrm{d}\mathbf{u} \tag{38}$$

Expression (38) suggests a derivation by cases approach to finding $\left.\frac{\partial P(\mathbf{d};\theta,\mathcal{N})}{\partial \gamma}\right|_{\gamma=0}$. Fortunately a brute force exhaustive approach is not required because it is possible to show that most of the summands in (38) do not influence the derivative at $\gamma = 0$.

Let $\tilde{\mathbb{B}}^n$ be the set of bucket configurations with at least two inner buckets. If at least two elements of $\mathbf{U}$ fall in inner buckets, then we have that $\mathbf{U} \in \mathbf{b}$ with $\mathbf{b} \in \tilde{\mathbb{B}}^n$. If, instead, at most one element of $\mathbf{U}$ falls in an inner bucket, then we have that $\mathbf{U} \in \mathbf{b}$ with $\mathbf{b} \in \mathbb{B}^n \setminus \tilde{\mathbb{B}}^n$. This set-up gives the likelihood decomposition:

$$P(\mathbf{d}; \theta, \mathcal{N}) = \tilde{P}(\mathbf{d}; \theta, \mathcal{N}) + Q(\mathbf{d}; \theta, \mathcal{N}), \tag{39}$$

with

$$\tilde{P}(\mathbf{d}; \theta, \mathcal{N}) = \sum_{\mathbf{b} \in \mathbb{B}^n \setminus \tilde{\mathbb{B}}^n} \int_{\mathbf{u} \in \mathbf{b}} \mathcal{N}(\mathbf{d}, \mathbf{u}; \theta) f_{\mathbf{U}}(\mathbf{u}) \, \mathrm{d}\mathbf{u} \tag{40}$$

$$Q(\mathbf{d}; \theta, \mathcal{N}) = \sum_{\mathbf{b} \in \tilde{\mathbb{B}}^n} \int_{\mathbf{u} \in \mathbf{b}} \mathcal{N}(\mathbf{d}, \mathbf{u}; \theta) f_{\mathbf{U}}(\mathbf{u}) \, \mathrm{d}\mathbf{u}. \tag{41}$$

To prove Theorem 2.3 we show that for $\gamma \to 0$

$$P(\mathbf{d}; \theta, \mathcal{N}) = \tilde{P}(\mathbf{d}; \theta, \mathcal{N}) + \mathcal{O}\left(\gamma^2\right). \tag{42}$$

Intuitively, this follows from the fact that the chance that two or more elements of $\mathbf{U}$ fall in inner buckets is negligible when γ is close to zero (because most of the probability mass for U_{ij} is contained in the two outer buckets when strategic interactions are small). Hence when calculating the optimal test statistic we are free to focus on the cases where either all, or all but one, of the elements of $\mathbf{U}$ fall in outer buckets. We can then show that

$$\left.\frac{\partial P\left(\mathbf{d};\theta,\mathcal{N}\right)}{\partial\gamma}\right|_{\gamma=0}=\left.\frac{\partial\tilde{P}\left(\mathbf{d};\theta,\mathcal{N}\right)}{\partial\gamma}\right|_{\gamma=0}. \tag{43}$$

Hence to derive the form of $\left.\frac{\partial P(\mathbf{d};\theta,\mathcal{N})}{\partial\gamma}\right|_{\gamma=0}$ we need only calculate $\left.\frac{\partial\tilde{P}(\mathbf{d};\theta,\mathcal{N})}{\partial\gamma}\right|_{\gamma=0}$. This calculation is non-trivial, but doable. Details of this calculation are provided in the proof.

Preliminary results

Lemma A.1. *Any differentiable function $f\in\mathcal{O}\left(\gamma^2\right)$ with $f\left(0\right)=0$ has a derivative of zero at point zero.*

Proof. For $f\in\mathcal{O}\left(\gamma^2\right)$ we have, for some $C>0$ and $\epsilon>0$, that

$$\left|f\left(\gamma\right)\right|<C\gamma^2 \tag{44}$$

for all $\gamma\in[-\epsilon,\epsilon]$. The derivative of f at $\gamma=0$ equals

$$f'\left(0\right)=\lim_{\gamma\to 0}\frac{f\left(\gamma\right)-f\left(0\right)}{\gamma}=\lim_{\gamma\to 0}\frac{f\left(\gamma\right)}{\gamma}, \tag{45}$$

with the second equality because $f\left(0\right)=0$. As $\gamma\to 0$, we will have $\gamma<\epsilon$ so that

$$f'\left(0\right)=\lim_{\gamma\to 0}\frac{f\left(\gamma\right)}{\gamma}\leq\lim_{\gamma\to 0}\frac{C\gamma^2}{\gamma}=\lim_{\gamma\to 0}C\gamma \tag{46}$$

which goes to zero as $\gamma\to 0$ as claimed. □

Proof of Theorem 2.3

We begin with the likelihood decomposition (38) given above. The number of summands in (38) depends on the partition that $s_{ij}(\mathbf{d})$ induces on $\mathbb{R}$. For a positive γ, the number neither depends on the exact value of γ, nor on the other covariates and parameters. Intuitively, as long as γ is positive, there is a positive probability that $\mathbf{U}$ falls in any combination of buckets. The number of summands in (38) is typically large. The buckets $\mathbf{b}$ of $\mathbb{B}^n$ and the function $\mathcal{N}$ depend on γ.

We have that

$$\begin{aligned}
\frac{\partial P(\mathbf{d};\theta,\mathcal{N})}{\partial\gamma} &= \frac{\partial}{\partial\gamma}\left\{\sum_{\mathbf{b}\in\mathbb{B}^n}\int_{\mathbf{u}\in\mathbf{b}}\mathcal{N}(\mathbf{d},\mathbf{u};\theta)\, f_{\mathbf{U}}(\mathbf{u})\,\mathrm{d}\mathbf{u}\right\}\\
&= \sum_{\mathbf{b}\in\mathbb{B}^n}\frac{\partial}{\partial\gamma}\int_{\mathbf{u}\in\mathbf{b}}\mathcal{N}(\mathbf{d},\mathbf{u};\theta)\, f_{\mathbf{U}}(\mathbf{u})\,\mathrm{d}\mathbf{u},
\end{aligned}$$

The switching of summation and derivative operator is possible because the number of summands does not depend on γ. We could try to take the derivative of each summands integral boundaries and of $\mathcal{N}(\mathbf{d},.;\theta)$. But there is no need to boil the ocean, because regardless of $\mathcal{N}(\mathbf{d},.;\theta)$ most of the summands are 0. To show this we consider three sets of summands.

Case 1: more than two buckets in B are inner buckets

Recall that the boldface subscripts $\mathbf{i}=\mathbf{1},\mathbf{2},\ldots$ index the $n=N(N-1)$ directed dyads in arbitrary order. Consider a set of buckets $\mathbf{b}$ where two or more of them are inner buckets. Without loss of generality assume that the $L\geq 2$ inner buckets correspond to $b_{\mathbf{1}},\ldots,b_{\mathbf{L}}$ of $\mathbf{b}=(b_{\mathbf{1}},\ldots,b_{\mathbf{n}})$. The shape of the $\mathbf{l}^{th}$ bucket is $(\gamma\underline{s}_{\mathbf{l}},\gamma\bar{s}_{\mathbf{l}}]$ with $\underline{s}_{\mathbf{l}}<\bar{s}_{\mathbf{l}}$ coinciding with the bucket borders induced by the precise form of strategic interaction specified under the alternative. We normalize the dyad-specific systematic utility component $\mu_{ij}=0$ without loss of generality.

Recall that $\tilde{\mathbb{B}}^n$ is the set of bucket configurations with two or more inner buckets. For any $\mathbf{b}\in\tilde{\mathbb{B}}^n$ we can derive the upper bound:

$$\begin{aligned}
\int_{\mathbf{u}\in\mathbf{b}}\mathcal{N}(\mathbf{d},\mathbf{u};\theta)\, f_{\mathbf{U}}(\mathbf{u})\,\mathrm{d}\mathbf{u} &= \int_{\mathbf{u}\in\mathbf{b}}\mathcal{N}(\mathbf{d},\mathbf{u};\theta)\left[\prod_{\mathbf{i}} f_U(u_{\mathbf{i}})\right]\mathrm{d}\mathbf{u}\\
&\leq \int_{\gamma\underline{s}_{\mathbf{1}}}^{\gamma\bar{s}_{\mathbf{1}}} f_U(u_{\mathbf{1}})\times\cdots\times\int_{\gamma\underline{s}_{\mathbf{L}}}^{\gamma\bar{s}_{\mathbf{L}}} f_U(u_{\mathbf{L}})\int_{\mathbf{u}_{-L}\in\mathbf{b}_{-L}} f_{\mathbf{U}_{-L}}(\mathbf{u}_{-L})\,\mathrm{d}\mathbf{u}\\
&< \int_{\gamma\underline{s}_{\mathbf{1}}}^{\gamma\bar{s}_{\mathbf{1}}} f_U(u_{\mathbf{1}})\times\cdots\times\int_{\gamma\underline{s}_{\mathbf{L}}}^{\gamma\bar{s}_{\mathbf{L}}} f_U(u_{\mathbf{L}})\,\mathrm{d}u_{\mathbf{1}}\cdots\mathrm{d}u_{\mathbf{L}}\\
&< \int_{\gamma\underline{s}_{\mathbf{1}}}^{\gamma\bar{s}_{\mathbf{1}}} 1\times\cdots\times\int_{\gamma\underline{s}_{\mathbf{L}}}^{\gamma\bar{s}_{\mathbf{L}}} 1\mathrm{d}u_{\mathbf{1}}\cdots\mathrm{d}u_{\mathbf{L}}\\
&= \gamma^L(\bar{s}_{\mathbf{1}}-\underline{s}_{\mathbf{1}})\times\cdots\times(\bar{s}_{\mathbf{L}}-\underline{s}_{\mathbf{L}})
\end{aligned}$$

where $\mathbf{u}_{-L}$ denotes the vector $\mathbf{u}$ after removal of its first L components and similarly for $\mathbf{b}_{-L}$. The first equality follows from independence of the components of $\mathbf{u}$, the second (weak) inequality from the fact that $\mathcal{N}(\mathbf{d},\mathbf{u};\theta)\leq 1$ for all $\mathbf{u}\in\mathbb{U}$. The third (strict) inequality follows because $f_{\mathbf{U}_{-L}}(\mathbf{u}_{-L})$ is a density and the integration is not over all of $\mathbb{R}^{n-L}$. The fourth (strict) inequality arises because when $f_U(u)$ is the logistic density we have that $f_U(u) = F_U(u)[1 - F_U(u)] < 1$ for all u on a compact interval of the real line. We conclude that any summand where $\mathbf{b}$ has two or more inner buckets is $\mathcal{O}(\gamma^2)$ for $\gamma \to 0$.

We have, directly from this argument, that $Q(\mathbf{d}; \theta, \mathcal{N}) \in \mathcal{O}(\gamma^2)$ and furthermore that $Q\left(\mathbf{d}; (0, \delta')', \mathcal{N}\right) = 0$ (since inner buckets have zero probability when $\gamma = 0$). Hence, by Lemma A.1, we have that

$$\left.\frac{\partial Q(\mathbf{d}; \theta, \mathcal{N})}{\partial \gamma}\right|_{\gamma=0} = 0.$$

This is enough to show equation (43) above. This simplification is essential to the overall result, as it allows us to proceed without knowing any details about the form of the equilibrium weight function $\mathcal{N}$ when $\mathbf{U}$ takes values which admit multiple NE networks.

Case 2: No bucket in b is an inner bucket (i.e., all buckets are outer buckets)

If all components of $\mathbf{u}$ fall in either their first or last buckets, then the network is uniquely defined. This occurs because agent-level preferences for forming (or not forming) a link are so strong that they do not depend on the presence or absence of other links in the network. Each agent i either prefers to send a link to j, regardless of the actions taken by others, or does not wish to send a link. Put differently, each agent has a pure link formation strategy which is strictly dominating in such games; therefore $\mathcal{N}(\mathbf{d}, \mathbf{u}; \theta)$ is either zero or one.

For a particular network $\mathbf{d}$, $\mathcal{N}(\mathbf{d}, \mathbf{u}; \theta) = 1$ if, for all (directed) dyads ij such that $d_{ij} = 1$, we have that u_{ij} falls in the first bucket and for all dyads ij such that $d_{ij} = 0$ we have that u_{ij} falls in the last bucket. These considerations give the equality

$$\int_{\mathbf{u}\in\mathbf{b}} \mathcal{N}(\mathbf{d}, \mathbf{u}; \theta) f_{\mathbf{U}}(\mathbf{u}) \,\mathrm{d}\mathbf{u} = \prod_{i\neq j} \left[\int_{-\infty}^{\mu_{ij}+\gamma\underline{s}} f_U(u_{ij}) \,\mathrm{d}u_{ij}\right]^{d_{ij}} \left[\int_{\mu_{ij}+\gamma\bar{s}}^{\infty} f_U(u_{ij}) \,\mathrm{d}u_{ij}\right]^{1-d_{ij}} \tag{47}$$

$$= \prod_{i\neq j} \left[F_U(\mu_{ij} + \gamma\underline{s})\right]^{d_{ij}} \left[1 - F_U(\mu_{ij} + \gamma\bar{s})\right]^{1-d_{ij}} \tag{48}$$

Taking logarithms of the expression above, differentiating with respect to γ, evaluating at $\gamma = 0$, and multiplying by $P_0(\mathbf{d}; \delta)$ yields a derivative for summands where all buckets in $\mathbf{b}$ are outer buckets of

$$P_0(\mathbf{d}; \delta) \sum_{i\neq j} \left[d_{ij}\underline{s}\frac{f_U(\mu_{ij})}{F_U(\mu_{ij})} - (1 - d_{ij})\,\bar{s}\frac{f_U(\mu_{ij})}{1 - F_U(\mu_{ij})}\right]. \tag{49}$$

Case 3: Exactly one bucket in b is an inner bucket

If all but one component of $\mathbf{u}$ falls into its first or last bucket, then the resulting network is uniquely defined except for the presence or absence of one arc, say, ij. For any such draw of $\mathbf{u}$, since all other links are formed according to a strictly dominating strategy, player i will either benefit from forming the ij arc or not. Hence $\mathcal{N}(\mathbf{d}, \mathbf{u}; \theta)$ is also either zero or one in this case.

For a particular network $\mathbf{d}$, $\mathcal{N}(\mathbf{d}, \mathbf{u}; \theta)$ will equal one if two conditions hold. First, for all directed dyads $kl \neq ij$ such that $d_{kl} = 1$ we have that u_{kl} falls in the first bucket and for all dyads $kl \neq ij$ such that $d_{kl} = 0$ we have that u_{kl} falls in the last bucket. Second, for the dyad ij with u_{ij} falling in an inner bucket, we require that if $u_{ij} \in [\mu_{ij} + \gamma \underline{s}, \mu_{kl} + \gamma s_{ij}(\mathbf{d}))$ that $d_{ij} = 1$, while if $u_{ij} = [\mu_{kl} + \gamma s_{ij}(\mathbf{d}), \mu_{ij} + \gamma \bar{s})$ we require that $d_{ij} = 0$. The overall likelihood contribution for this case therefore equals:

$$
\begin{aligned}
\int_{\mathbf{u}\in\mathbf{b}} \mathcal{N}(\mathbf{d}, \mathbf{u}; \theta) f_{\mathbf{U}}(\mathbf{u})\, \mathrm{d}\mathbf{u} = & \prod_{kl\neq ij} \left[\int_{-\infty}^{\mu_{kl}+\gamma\underline{s}} f_U(u_{kl})\, \mathrm{d}u_{kl}\right]^{d_{kl}} \left[\int_{\mu_{kl}+\gamma\bar{s}}^{\infty} f_U(u_{kl})\, \mathrm{d}u_{kl}\right]^{1-d_{kl}} \\
& \times \left[\int_{\mu_{ij}+\gamma\underline{s}}^{\mu_{ij}+\gamma s_{ij}(\mathbf{d})} f_U(u_{ij})\, \mathrm{d}u_{ij}\right]^{d_{ij}} \left[\int_{\mu_{ij}+\gamma s_{ij}(\mathbf{d})}^{\mu_{ij}+\gamma\bar{s}} f_U(u_{ij})\, \mathrm{d}u_{ij}\right]^{1-d_{ij}} \\
= & \prod_{kl\neq ij} [F_U(\mu_{kl} + \gamma\underline{s})]^{d_{kl}} [1 - F_U(\mu_{kl} + \gamma\bar{s})]^{1-d_{kl}} \\
& \times [F_U(\mu_{ij} + \gamma s_{ij}(\mathbf{d})) - F_U(\mu_{ij} + \gamma\underline{s})]^{d_{ij}} \\
& \times [F_U(\mu_{ij} + \gamma\bar{s}) - F_U(\mu_{ij} + \gamma s_{ij}(\mathbf{d}))]^{1-d_{ij}}.
\end{aligned}
$$

Recall that $s_{ij}(\mathbf{d})$ is the "strategic" part of the marginal utility agent i gets if he forms ij. Because the last two terms in $[\cdot]$ in the expression above are zero at $\gamma = 0$ we only need to consider their derivative (by the product rule the other term equals zero at $\gamma = 0$). Differentiating the last two terms with respect to γ (and multiplying by the balance of preceding terms) yields

$$
\begin{aligned}
&\prod_{kl\neq ij}\left[F_U\left(\mu_{kl}+\gamma\underline{s}\right)\right]^{d_{kl}}\left[1-F_U\left(\mu_{kl}+\gamma\bar{s}\right)\right]^{1-d_{kl}}\\
&\times\left[s_{ij}\left(\mathbf{d}\right)f_U\left(\mu_{ij}+\gamma s_{ij}\left(\mathbf{d}\right)\right)-\underline{s}f_U\left(\mu_{ij}+\gamma\underline{s}\right)\right]^{d_{ij}}\\
&\times\left[\bar{s}f_U\left(\mu_{ij}+\gamma\bar{s}\right)-s_{ij}\left(\mathbf{d}+ij\right)f_U\left(\mu_{ij}+\gamma s_{ij}\left(\mathbf{d}+ij\right)\right)\right]^{1-d_{ij}}\\
=&\prod_{i\neq j}\left[F_U\left(\mu_{ij}+\gamma\underline{s}\right)\right]^{d_{ij}}\left[1-F_U\left(\mu_{ij}+\gamma\bar{s}\right)\right]^{1-d_{ij}}\\
&\times\left[s_{ij}\left(\mathbf{d}\right)\frac{f_U\left(\mu_{ij}+\gamma s_{ij}\left(\mathbf{d}\right)\right)}{F_U\left(\mu_{ij}+\gamma\underline{s}\right)}-\underline{s}\frac{f_U\left(\mu_{ij}+\gamma\underline{s}\right)}{F_U\left(\mu_{ij}+\gamma\underline{s}\right)}\right]^{d_{ij}}\\
&\times\left[\bar{s}\frac{f_U\left(\mu_{ij}+\gamma\bar{s}\right)}{F_U\left(\mu_{ij}+\gamma\bar{s}\right)}-s_{ij}\left(\mathbf{d}\right)\frac{f_U\left(\mu_{ij}+\gamma s_{ij}\left(\mathbf{d}\right)\right)}{F_U\left(\mu_{ij}+\gamma\bar{s}\right)}\right]^{1-d_{ij}}.
\end{aligned}
$$

Summing this expression over all potential arcs (and evaluating at $\gamma=0$) gives a total contribution of "one inner bucket in $\mathbf{b}$" summands to the derivative of:

$$
\begin{aligned}
P_0\left(\mathbf{d};\delta\right)\sum_{i\neq j}d_{ij}&\left[s_{ij}\left(\mathbf{d}\right)\frac{f_U\left(\mu_{ij}\right)}{F_U\left(\mu_{ij}\right)}-\underline{s}\frac{f_U\left(\mu_{ij}\right)}{F_U\left(\mu_{ij}\right)}\right]\\
&+\left(1-d_{ij}\right)\left[\bar{s}\frac{f_U\left(\mu_{ij}\right)}{F_U\left(\mu_{ij}\right)}-s_{ij}\left(\mathbf{d}\right)\frac{f_U\left(\mu_{ij}\right)}{F_U\left(\mu_{ij}\right)}\right].
\end{aligned}
\tag{50}
$$

Summing (47) and (50) then gives the expression in the statement of Theorem 2.3. Using similar methods we can show that $P\left(\mathbf{d};\theta\right)$ can be differentiated with respect to γ twice as claimed.

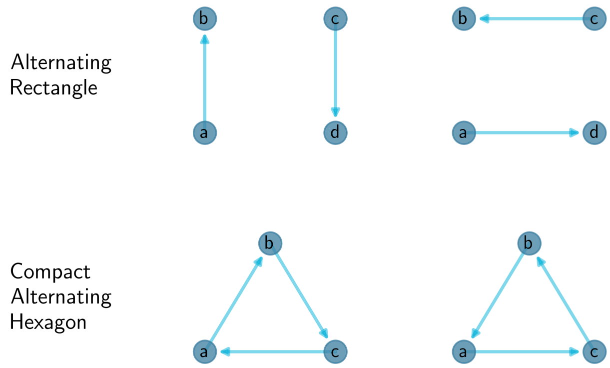
A.4 MCMC Proofs

Before proving Theorem 3.1, we provide some additional discussion of our MCMC algorithm. This also allows us to introduce additional notation required for the proof.

Our algorithm involves finding sequences of schlaufen such that when we "swap their edges" the in-degree and out-degree sequence of the network remains unchanged. Individually, such swaps may induce cross link matrix violations, but by swapping the edges in many schlaufen simultaneously, we can ensure such violations net to zero.

First consider edge swaps which leave the in- and out-degree sequence of a network unchanged. Figure 5 depicts two canonical alternating cycles (schlaufen) (Rao et al., 1996). The first, $C:=abcda$, is a so called *alternating rectangle*. In the configuration to the left a and d each have a single out-link and b and c a single in-link; this is also true in the configuration to the right, which corresponds to the the network generated by switching C. The second

Figure 5: Examples of alternating cycles



	a	b	c	d	a
i	1	2	3	4	5
sel. Pr	$\frac{1}{4}$	$\frac{1}{1}$	$\frac{1}{3}$	$\frac{1}{1}$	$\frac{1}{2}$

$\Pr(R_1) = \frac{1}{24}$

	a	b	c	a	b	c	a
i	1	2	3	4	5	6	7
sel. Pr	$\frac{1}{3}$	$\frac{1}{1}$	$\frac{1}{1}$	$\frac{1}{1}$	$\frac{1}{1}$	$\frac{1}{1}$	$\frac{1}{1}$

$\Pr(R_1) = \frac{1}{3}$

Source: Authors' calculations.
Notes: The first row depicts an alternating rectangle before and after switching. The second row depicts a compact alternating hexagon before and after switching. The final column of the figure shows the probability with which each node is chosen, resulting in the total probability of the schlaufe.

cycle, called a *compact alternating hexagon*, can be constructed from a so-called $030C$ triad.

Let $\mathbb{G}_\mathbf{s}$ denote the set of all digraphs with degree sequence $\mathbf{s}$. Rao et al. (1996) showed that, for $\mathbf{G}$ and $\mathbf{G}'$ distinct and belonging to $\mathbb{G}_\mathbf{s}$, it is possible to obtain $\mathbf{G}'$ from $\mathbf{G}$ by switching a sequence of alternating rectangles and compact alternating hexagons. In practice it may be useful, in the sense that one can move from $\mathbf{G}$ to $\mathbf{G}'$ with fewer cycle switches, if longer alternating cycles are used (e.g., McDonald et al., 2007; Tao, 2016).

These concepts are further illustrated in Figure 6, which also illustrates the important idea of the cross-link violation matrices, associated with a sequence of schlaufen, summing to zero. Consider the network depicted in Panel A of Figure 6. This network consists of two types of agents: gold (light) and blue (dark). The cross link matrix for the graph is given in Panel D. In Panels B and C a sequence of three schlaufen is shown. The first schlaufen is $R_1 = jgabcdeca$. It is constructed through a sequence of active and passive steps described in the main text (see also the notes to Figure 2 in the main text). We begin by choosing agent j randomly with a probability of $\frac{1}{10}$ (since there are ten agents in the network). We then take an active step, randomly choosing one of the two agents to which j directs a link (i.e., either agent g or i). Here we choose agent g. Next we take a passive step. Specifically we choose an agent at random from the set of agents that *do not* direct a link to g (the agent chosen in the previous active step). The probability associated with our choice in this passive step is $\frac{1}{7}$; this corresponds to the reciprocal number of agents in the network (i.e., 10) minus the indegree the current agent (i.e., 2) minus one (since self-loops are not allowed). We continue taking active and passive steps in this way until we visit a for the second time. At this point we stop since our schlaufe now includes the alternating cycle $C_1 = abcdeca$. Note that c is also visited twice, but also that $cdec$ is not an alternating cycle since it is not of even length (see Definition 3.2).

As seen in the example we can calculate the probability of a schlaufe R as we go through the algorithm (see Panel E). In Step 1 of Algorithm 2 an agent is chosen with probability $\frac{1}{N}$. Next let $r^a_D(i)$ be the cardinality of the set of feasible out links in an active step. This set consists of all the out links of node i, which are not already marked in $\mathbf{D}$. Similarly, let $r^p_D(i)$ be the cardinality of the set of feasible outlinks in an passive step. That set consists of all the links ij for which ji is not in $\mathbf{D}$ and which are not already marked. The probability of $R = (i_1, .., i_l)$ can now be written as

$$p_{\mathbf{D}}(R) = \frac{1}{N} \prod_{k=1}^{l-1} \left(\frac{1}{r^a_{\mathbf{D}}(i_k)} [k \bmod 2] + \frac{1}{r^p_{\mathbf{D}}(i_k)} [(k-1) \bmod 2] \right) \tag{51}$$

In step 2 of Algorithm 1 we attempt to find a sequence of schlaufen with probability $1 - q$ and do not change the adjacency matrix otherwise. In step 3, a schlaufen sequence $\mathcal{R} = (R_1, .., R_h)$ is constructed/found. After each detected schlaufe in this sequence, say R_k, any cycle in it is marked. Let $\mathbf{D}_k$ be the graph with the cycles of $R_1, .., R_{k-1}$ marked. After each schlaufe added the construction is stopped with probability $\frac{1}{2}$. The probability of finding a cycle R_k is $p_{\mathbf{D}_k}(R_k)$ as given in equation (51) above. The total probability of a feasible schlaufen sequence $\mathcal{R}$ is therefore

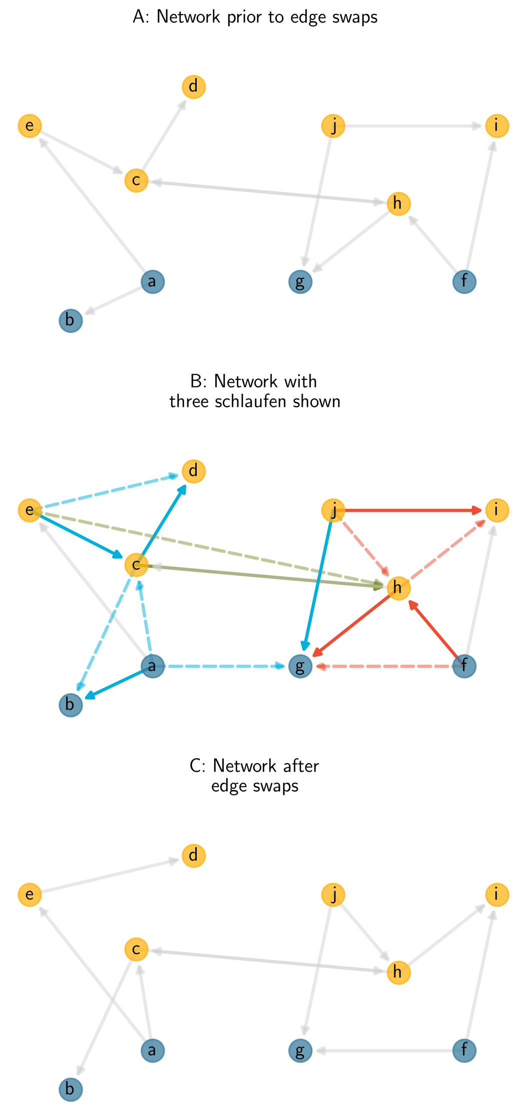
$$p_{\mathbf{D}}(\mathcal{R}) = (1 - q) \frac{1}{2^{(h-1)}} \prod_{i=1}^{h} p_{\mathbf{D}_k}(R_k). \tag{52}$$

To show that our algorithm does indeed generate a uniform random draw from the set $\mathbb{D}_{\mathbf{s},\mathbf{m}}$ we use standard Markov chain theory (e.g., Chapters 7 and 10 of Mitzenmacher and Upfal (2005)).

The random rewiring of the network implemented by Algorithm 1 can be described as a Markov chain. To show that, for τ large enough, it returns a uniform random draw from $\mathbb{D}_{\mathbf{s},\mathbf{m}}$, we prove that the stationary distribution of the Markov chain generated by Algorithm 1 is uniform on $\mathbb{D}_{\mathbf{s},\mathbf{m}}$. To show this it is helpful to develop a graphical representation of the Markov chain.

We denote the state graph of the Markov chain by $\Phi = (\mathcal{V}_\phi, \mathcal{A}_\phi)$. Its underlying vertex set $\mathcal{V}_\phi$ is the set of all elements in $\mathbb{D}_{\mathbf{s},\mathbf{m}}$: each node in our state graph is a network with degree

Figure 6: A feasible schlaufen sequence



D: Cross-link (PAM) matrix

	Blue	Gold
Blue	1	3
Gold	2	5

E: Three schlaufen

	j	g	a	b	c	d	e	c	a
i	1	2	3	4	5	6	7	8	9
sel. Pr	$\frac{1}{10}$	$\frac{1}{2}$	$\frac{1}{7}$	$\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{2}$	$\frac{1}{8}$	$\frac{1}{1}$	$\frac{1}{8}$

	c	h	e
i	1	2	3
sel. Pr	$\frac{1}{10}$	$\frac{1}{1}$	$\frac{1}{7}$

	f	h	j	i	h	g	f
i	1	2	3	4	5	6	7
sel. Pr	$\frac{1}{10}$	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{1}$	$\frac{1}{7}$	$\frac{1}{1}$	$\frac{1}{6}$

F: Violation matrices for the three schlaufen

	Blue	Gold
Blue	-1	+1
Gold	+1	-1

	Blue	Gold
Blue	0	0
Gold	0	0

	Blue	Gold
Blue	+1	-1
Gold	-1	+1

Source: Authors' calculations.
Notes: See the discussion in the text. The figure depicts three link disjoint schlaufen with violation matrices which sum to zero. Panel E reports the (ex ante) probability that a given node was selected as the schlaufe was constructed. See equation (51).

sequence $\mathbf{S} = \mathbf{s}$ and cross link matrix $\mathbf{M} = \mathbf{m}$. For network $\mathbf{D}$ in $\mathbb{D}_{\mathbf{s},\mathbf{m}}$, we denote by $v_{\mathbf{D}}$ the corresponding vertex in $\mathcal{V}_\phi$. The arc set $\mathcal{A}_\phi$ is defined as follows.

1. For all vertices we add the self loop $(v_{\mathbf{D}}, v_{\mathbf{D}})$ with (probability) weight q (see Step 2 of Algorithm 1).

2. Let $\mathbf{D}$ and $\mathbf{D}'$ be two different networks in $\mathbb{D}_{\mathbf{s},\mathbf{m}}$. Let $\mathbf{D}\Delta\mathbf{D}'$ equal the union of the set of edges in $\mathbf{D}$, but not in $\mathbf{D}'$ and the set of edges in $\mathbf{D}'$, but not in $\mathbf{D}$. For each feasible schlaufen-sequence $\mathcal{R}$, with cycle edge set equal to $\mathbf{D}\Delta\mathbf{D}'$ we add the edge $(v_{\mathbf{D}}, v_{\mathbf{D}'})$ and assign to it probability weight $p_{\mathbf{D}}(\mathcal{R})$.

3. Finally we add a directed loop $(v_{\mathbf{D}}, v_{\mathbf{D}})$ if the probability of all arrows leaving $v_{\mathbf{D}}$, introduced in points 1 and 2 immediately above, do not sum to 1. The probability of this loop is 1 minus the sum of the probability of all other outward arrows.

The probability of any arc $a \in \mathcal{A}_\phi$ is denoted by $p(a)$. Note, by definition, the state graph can have parallel arcs and loops.

With these definitions in place we can prove correctness of the algorithm. First we show that the probability of the algorithm moving from graph $\mathbf{D}$ to $\mathbf{D}'$ coincides with the probability of moving in the reverse direction.

Lemma A.2. *For any two vertexes $v_{\mathbf{D}}, v_{\mathbf{D}'}$ the transition probability attached to $(v_{\mathbf{D}}, v_{\mathbf{D}'})$ equals that attached to $(v_{\mathbf{D}'}, v_{\mathbf{D}})$.*

Proof. Let $A_{\mathbf{D},\mathbf{D}'}$ be the set of arcs form the node $v_{\mathbf{D}}$ to the node $v_{\mathbf{D}'}$. We construct a bijection $\varphi : A_{\mathbf{D},\mathbf{D}'} \to A_{\mathbf{D}',\mathbf{D}}$. Then we show that the probability of an arc $p(a)$ is equal to $p(\varphi(a))$. If that is proven, the probability of a transition form $v_{\mathbf{D}}$ to $v_{\mathbf{D}'}$ is

$$\begin{aligned}
\sum_{a \in A_{\mathbf{D}',\mathbf{D}}} p(a) &= \sum_{a \in A_{\mathbf{D}',\mathbf{D}}} p(\varphi(a)) \\
&= \sum_{\varphi^{-1}(a') \in A_{\mathbf{D}',\mathbf{D}}} p(a') \\
&= \sum_{a' \in \varphi(A_{\mathbf{D}',\mathbf{D}})} p(a') \\
&= \sum_{a' \in A_{\mathbf{D},\mathbf{D}'}} p(a')
\end{aligned}$$

which is the probability for a transition from $v_{\mathbf{D}'}$ to $v_{\mathbf{D}}$.

For the construction of the bijection consider that every arc $A_{\mathbf{D},\mathbf{D}'}$ corresponds uniquely to a schlaufen-sequence $\mathcal{R} = (R_1, .., R_h)$. Let $R_k = (i_1, .., i_m, .., i_l)$ with i_m the start of the cycle (if there is no cycle in R, we set $R = \bar{R}$). We define $\bar{R}_k = (i_1, .., i_m, i_{l-1}, ..i_{m+1}, i_l)$ and $\bar{\mathcal{R}} = (\bar{R}_1, .., \bar{R}_h)$.

Note that the $R_1, .., R_h$ are link disjoint and as soon as the cycle of R_k is switched $\bar{R}_k$ is a schlaufe. The violation matrix of $\bar{R}_k$ is the negative violation matrix of R_k. This implies that if $\mathcal{R}$ is a feasible schlaufen sequence for G which defined an arc in $A_{\mathbf{D},\mathbf{D}'}$ then $\bar{\mathcal{R}}$ is a feasible schlaufen-sequence for $\mathbf{D}'$ and defines an arc $A_{\mathbf{D}',\mathbf{D}}$.

We define now φ as the function which maps the arc in $A_{\mathbf{D},\mathbf{D}'}$ with schlaufen sequence $\mathcal{R}$ to the arc in $A_{\mathbf{D}',\mathbf{D}}$ with schlaufen sequence $\bar{\mathcal{R}}$. By construction φ is injective, which implies $|A_{\mathbf{D},\mathbf{D}'}| \leq |A_{\mathbf{D}',\mathbf{D}}|$. By symmetry we conclude $|A_{\mathbf{D}',\mathbf{D}}| \geq |A_{\mathbf{D},\mathbf{D}'}|$, which implies $|A_{\mathbf{D}',\mathbf{D}}| = |A_{\mathbf{D},\mathbf{D}'}|$ and that φ is bijective.

It remains to show that the probability of an arc $p(a)$ is equal to $p(\varphi(a))$. For any node there are equally many feasible active / passive outlinks in $\mathbf{D}$ as in $\mathbf{D}'$. If for a node one outlink is marked due to an link in R_k then for the same node one outlink is marked in $\bar{R}_k$. Therefore $r_{G'_k}(i)$ is equal to $r_{G_k}(i)$ for an active as well as a passive step. Looking at equation (51) the $p_{G_k}(R_k)$ is only different from $p_{G'_k}(\bar{R}_k)$ in the numbering of the factors. But in a cycle of a schlaufe the start node i_m and the end node i_l are such that $m - l \mod 2 = 0$. The reordering leaves even indexes even and odd indexes odd. Therefore $p_G(R_k) = p'_G(\bar{R}_k)$. From equation (52) it follows directly that $p_G(\mathcal{R}) = p_{G'}(\bar{\mathcal{R}})$ which completes the proof.

□

Next we show the state graph is strongly connected. This means our Algorithm moves from any $\mathbf{D} \in \mathbb{D}_{\mathbf{s},\mathbf{m}}$ to any other $\mathbf{D}' \in \mathbb{D}_{\mathbf{s},\mathbf{m}}$ with positive probability.

Lemma A.3. *The state graph Φ is strongly connected.*

Proof. The symmetric difference of two realizations of $\mathbb{D}_{s,m}$, which we denote by $\mathbf{D}$ and $\mathbf{D}'$ is a set of alternating cycles. Cycles are in particular schlaufen. We order them arbitrarily as $(R_1, .., R_h)$. The sum of the violation matrices is 0. Therefore $(R_1, .., R_h)$ is either a feasible schlaufen-sequence or a concatenation of feasible schlaufen-sequences. In the first case there is an arc from $v_{\mathbf{D}}$ to $v_{\mathbf{D}'}$. In the second case, all the feasible schlaufen-sequence define an arc to a new node, resulting in a directed path starting at $v_{\mathbf{D}}$ and ending in $v_{\mathbf{D}'}$. Thus between any two vertexes in Φ there is a directed path. □

With Lemmata A.2 and A.3 in hand, proving Theorem 3.1 is straightforward.

Proof of Theorem 3.1

Every time the Algorithm 1 arrives at step 2 a new arc of the state graph is crossed. At step 2 the algorithm follows a loop arc of type 1 with probability q. Otherwise it proceeds to step 3. In step 3 a schlaufen-sequence $\mathcal{R}$ is constructed. If the violation matrices of this schlaufen-sequence sum up to 0, the its cycles are switched and an arc of type 2 is followed with probability $p_G(\mathcal{R})$. If the violation matrices do not sum up to 0, then an arc of type 3 is followed. All the cases in which the violation matrices don't sum up to 0 correspond to the residual probability. Therefore Algorithm 1 is a a random walk on the state graph Φ.

According to Lemma A.2 Φ is (weighted) symmetric and according to Lemma A.3 it is strongly connected. Due to the self-loops, Φ is not bipartite. Therefore the limit distribution is uniform.

A.5 Monte Carlo experiments

In this appendix we summarize the results of a small number of Monte Carlo experiments. These experiments illustrate the excellent size control and good power properties of our tests. For the Monte Carlo experiments we work with the general utility function introduced in the main paper. We assume that $A_i \in \mathbb{A} \overset{def}{\equiv} \{\alpha_L, \alpha_H\}$, $B_i \in \mathbb{B} \overset{def}{\equiv} \{\beta_L, \beta_H\}$ and $X_i \in \mathbb{X} \overset{def}{\equiv} \{0, 1\}$. We assume that each support point in $\mathbb{A} \times \mathbb{B} \times \mathbb{X}$ occurs with equal probability (i.e., with probability equal to $\frac{1}{8}$).

Observe that their are four types of sending agents: $(A_i = \alpha_L, X_i = 0)$, $(A_i = \alpha_H, X_i = 0)$, $(A_i = \alpha_L, X_i = 1)$ and $(A_i = \alpha_H, X_i = 1)$. Similarly there are four types of receiving agents. The null model is therefore fully described by $16 = 4 \times 4$ linking probabilities. These probabilities are, in turn, a function of the 8 model parameters. We set these parameters as follows: $-\alpha_L = \alpha_H = 0.7$, $-\beta_L = \beta_H = 0.5$, $\lambda_{00} = \lambda_{11} = -2$ and $\lambda_{10} = \lambda_{01} = -4$. This yields a null model expected graph density of about 0.10.

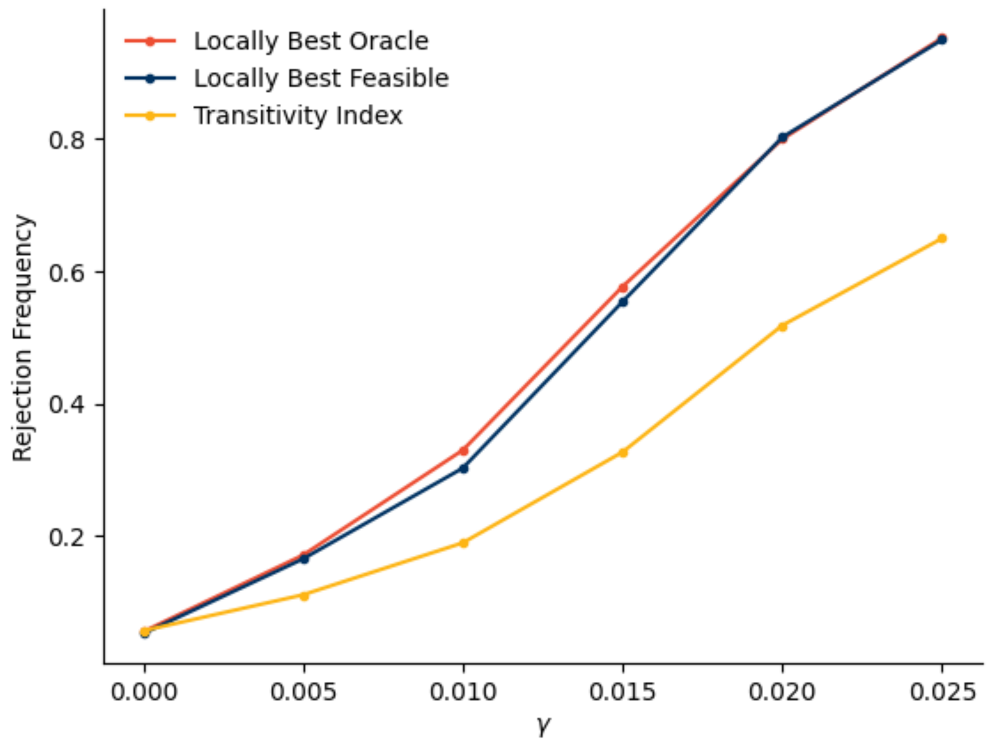
Our parameter choices generate meaningful degree heterogeneity and homophily under the null. Across $1,000$ Monte Carlo simulations with $N = 100$, average network density was 0.1, average reciprocity was 0.1, and the *average* standard deviation of in- and out-degree, was 7.5.

We set the network benefit function to $g_i(\mathbf{d}) = \sum_j d_{ij} \left(\sum_k d_{ik} d_{kj}\right)$ as is appropriate when agents prefer transitive ties. To simulate a network under the alternative we draw $\mathbf{U}$ and then, starting with an empty adjacency matrix, iterate until all links with positive marginal utility are present, and all those with negative marginal utility are absent. By Tarski's Theorem this finds us the least dense pure strategy Nash Equilibrium.

We compare the performance of three tests: (i) the infeasible locally best test that is based upon the true value of $\delta = \delta_0$; (ii) the feasible version of this test which replaces δ with its maximum likelihood estimate computed under the null; finally, (iii) we construct an *ad hoc* test based upon the transitivity index. This last test is the one most often used in practice (where it is compared to a reference value derived from a simple random graph null).

Figure 7 summarizes our findings. The horizontal axis of the figure correspond to different values of the strategic interaction parameter, γ_0; the vertical axis to the rejection frequency. With 1000 Monte Carlo replications the standard error of our simulation estimate of size is $\sqrt{(0.05\,(1-0.05)\,/1000)} \approx 0.007$.

Figure 7: Power Analysis



Source: Authors' calculations.
Notes: The figures plot the frequency with which $H_0 : \gamma_0 = 0$ is rejected across $1,000$ Monte Carlo replications for networks with $N = 100$ agents. The y-axis reports the estimated rejection frequency, the x-axis gives the value of the strategic interaction parameter, γ. The minimal pure strategy NE is use to simulate each network. For each simulation a total of 100 MCMC draws from $\mathbb{D}_{\mathbf{s},\mathbf{m}}$ were used to compute critical values. The mixing time was chosen such that (approximately) every edge is twice randomly modified before the network is considered a uniform draw. The marginal utility function equals $A_i + B_j + X_i'\Lambda_0 X_j + \gamma_0 s_{ij}(\mathbf{d}) - U_{ij}$ with $s_{ij}(\mathbf{d}) = \sum_k d_{ik}d_{kj} + \sum_{k \neq j} d_{ik}d_{jk}$. The distribution of (A_i, B_i, X_i) and the model parameters are as described in the main text.

As expected, the actual size of our test is indistinguishable (i.e., equal up to simulation error) from its nominal size. For the designs considered here the power gains associated with using the locally best test statistic derived in Section 2 are considerable. Furthermore the feasible locally best test, which replaces δ_0 with its MLE (computed under the null), performs almost as well as the infeasible locally best test based on the actual value of δ_0.

The Monte Carlo experiments highlight that the locally best test, which upweights episodes of "unexpected" transitivity, is more powerful than the ad hoc test based on comparing the transitivity index with its null distribution. Note both tests are valid and correctly-sized.

Next we consider the behavior of our test under mis-specification. Specifically we consider a data generating process where the link-specific random utility shocks are Gaussian instead of logistic. We set the variance of the Gaussian distribution to $\pi^2/3$ so that its scale is the same as in the logistic case. All other model parameters remain as defined above. We simulate $1,000$ networks with Gaussian errors, but then proceed "as if" our logistic assumption were true. Note our sufficiency, conditioning, and similarity arguments are no longer valid.

The results of these experiments are summarized in Figure 8. Our tests are conservative in these experiments, with actual sizes below their nominal 0.05 level. The power properties of our tests remain good. Note the "oracle" test is based on an evaluation of a logit probability function at the true utility function coefficients (from the the Gaussian model). The feasible test is based upon quasi maximum likelihood estimates of the pseudo-true utility function coefficients (under the null). There is no a priori reason to expect one test to perform better than the other in these designs (indeed one might expect the feasible test to do better under misspecification). In this example the two power curves cross.

This is just one experiment, but it provides some suggestive evidence that our test may still be useful in settings with modest departures from the logistic assumption.

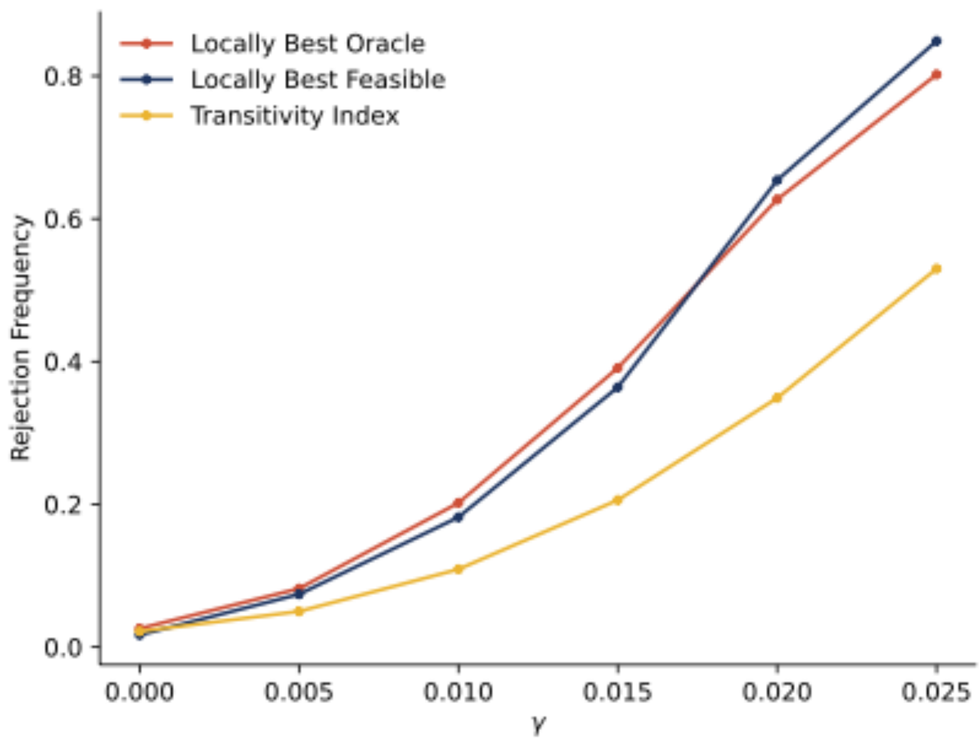
A.5.1 A note on computation

In our simulation experiments $\hat{\delta}$ only enters the locally best test statistic through the null linking probabilities $P(\hat{\mu}_{ij})$. This is relevant because in some settings the MLE of δ_0 may not exist, while that of $P(\mu_{ij})$ does (c.f., Chamberlain, 1980). This can be especially true if $\mathbf{D}$ is "sparse".

Consider a simple case where one agent sends no links in $\mathbf{D}$. The MLE of her out-degree parameter will be negative infinity. The MLEs of her dyadic out-link probabilities will all equal zero (which is finite). If we consider the form of the optimal test statistic, we see that any dyads ij, where i sends no links in the network in hand, makes a contribution of zero to the feasible test statistic. An analogous argument can be used to show that any ego which directs a link to all alters will also not contribute to the locally best test statistic.

In general, the MLEs of the baseline model choice probabilities are well-defined, although in a "sparse" graph some of these fitted probabilities may be zero or one. In such cases these

Figure 8: Test Behavior Under a Gaussian Random Utility Distribution



Source: Authors' calculations.
Notes: All features of these experiments are as reported in the notes to Figure 7 above, with the exception that the link-specific random utility shocks are Gaussian (with a variance of $\pi^2/3$). Estimation proceeds "as if" the logistic assumption nevertheless holds.

dyads simply make no contribution to the test statistic. Our code automatically handles such cases.

Recall that our test remains admissible for any $\delta \in \Delta$.

A.6 Additional applications

Bi-partite networks

Let $i = 1, \ldots, N_1$ index a set of firms deciding which of $j = 1, \ldots, N_2$ markets to enter or not. To be concrete consider the problem of airlines deciding which routes they will operate in. The $K_1 \times 1$ binary vector X_i^f indicates what type firm i is (e.g., a legacy carrier or a low cost airline); the binary vector X_j^m indicates what type market j is (e.g., small, medium or large). Let $\mathbf{E} = [E_{ij}]_{i=1,\ldots,N_1, j=1,\ldots,N_2}$ be the $N_1 \times N_2$ matrix which records which markets each firm chooses to enter ("$\mathbf{E}$" for market entry matrix).

The payoff firm $i = 1, \ldots, N_1$ gets from a given industry-wide pattern of market entry $\mathbf{E} = \mathbf{e}$ equals

$$\nu_i\left(\mathbf{e}_i, \mathbf{e}_{-i}; \theta, \mathbf{U}_i\right) = \gamma_0 g_i\left(\mathbf{e}\right) + \sum_{j=1}^{N_2} e_{ij}\left(A_i + B_j + X_i^{f\prime} \Delta X_j^m - U_{ij}\right). \tag{53}$$

Here A_i captures unobserved heterogeneity across firms; for example some firms may be able to operate at systematically lower cost and hence profitably enter more markets (corresponding to high values of A_i). Likewise B_j captures heterogeneity across markets; some markets may be intrinsically more profitable than others and hence many firms operate in them (corresponding to high values of B_j). The term $X_i^{f\prime} \Delta X_j^m$ allows certain types of markets to be systematically more attractive to certain types of firms. Finally U_{ij} is a firm-by-market specific logistic profit shock.

In a simple entry game, we might set $g_i\left(\mathbf{e}\right) = -\sum_{j=1}^{N_2} e_{ij}\left[\sum_{k=1}^{N_1} e_{kj}\right]$, such that entry into market j is less attractive to firm i when many other firms k also enter market j (e.g., Ciliberto and Tamer, 2009). This payoff function implies independence of entry decisions across markets.

An advantage of thinking about entry decisions as resulting in a bi-partite network, with arcs from firms to markets, is that it makes it easy to consider more complex preference structures. These preference structures can allow for interdependence in entry decisions across markets.

For example, in the model of Jia (2008) entry into market j is more attractive if the firm also enters other nearby markets. In this case we might set

$$g_i(\mathbf{e}) = -\sum_{j=1}^{N_2} e_{ij} \left[\sum_{k=1}^{N_1} e_{kj} \right] + \lambda(\mathbf{e}_i) \tag{54}$$

where $\lambda(\mathbf{e}_i)$ varies inversely with some measure of the spatial spread of those markets i enters (calibrated to measure how the operating costs of the firm vary with the geographic dispersion of the markets entered). In the airline example it may be more costly, or less profitable, to operate across markets that are disconnected. That is an airline may prefer route maps which allow a customer to travel to all other airports in their network without having to fly with competitor. In this example $\lambda(\mathbf{e}_i)$ might equal a decreasing function of the number of connected components in i's route network. Alternatively it could vary with the number of location pairs which can be reached on either a direct flight or by making just one connection.

We can also allow for more complex forms of spatial competition. If j indexes airline routes, then we might set

$$g_i(\mathbf{e}) = -\sum_{j=1}^{N_2} e_{ij}\phi_j(e_{-i}) \tag{55}$$

with $\phi_j(\mathbf{e}_{-i})$ returning how many competitor airlines operate in routes with origin and destination airports both within a one hour drive of the corresponding route j airports (e.g., entry into the SFO-LAX market may depend on how many competitors operate on the OAK-BUR, SFO-BUR, and OAK-LAX routes as well as the number which operate on the SFO-LAX route). One apparent advantage of this "network" perspective is that it allows for the incorporation of complex cross-market complementarities as well as rich forms of spatial competition.

To connect this entry model to the directed network problem defined in the main text we define the $N \times N$ matrix:

$$\mathbf{D} = \begin{bmatrix} \mathbf{0} & \mathbf{E} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \tag{56}$$

and then proceed as described in the paper.

The network $\mathbf{D}$ is bipartite. There are no firm-to-firm or market-to-market links. Furthermore, only firms may direct links (with arcs denoting market entry decisions). These features of the problem induce the special structure of $\mathbf{D}$ above. The cross link matrix will also have a structure analogous to that of the adjacency matrix.

In this example the null reference distribution will assign a zero to many "links" with probability one; this is not a problem for our MCMC simulation algorithm. The implementation available in the Python **ugd** package can handle degree sequences and cross link matrices with zero elements.

The null set of networks corresponds to all networks where the same number of firms enter each market as observed in the network in hand and the observed patterns of entry are the same as in the network in hand. For example, the number of low cost airlines entering small, medium and large markets is held fixed and so on.

A natural test statistic would be

$$R(\mathbf{D}) = \sum_{i=1}^{N_1} \sum_{j=1}^{N_2} \left(E_{ij} - p_{ij}\left(\hat{\delta}\right) \right) s_{ij}(E_{ij}) \tag{57}$$

with $p_{ij}\left(\hat{\delta}\right)$ an estimate of the probability that firm i enters market j under the null that $\gamma_0 = 0$ and $s_{ij}(\mathbf{E})$ the marginal network benefit of entering market j for firm i holding all other own and competitor route choices fixed.

As the above example makes clear, the application of the methods proposed in the main text to bipartite networks is conceptually straightforward. This, in turn, expands the class of many player games to which our methods apply. Note that questions of test power are game specific.

Conditional inference in dyadic models

Following a suggestion in Graham (2017), our MCMC algorithm can also be used for conditional maximum likelihood estimation (CMLE) of non-strategic dyadic logit models (see also Bartolucci (2024)). Let $D_{ij} = 1$ if country i attacks country j within some researcher-defined time period. Let X_i be a $K \times 1$ vector of country types (e.g., a partition of countries into broad geographic regions), finally let $Z_{ij} = 1$ if both i and j are democracies and zero otherwise. We posit the following model for the initiation of conflict by i against j

$$D_{ij} = \mathbf{1}\left(A_i + B_j + X_i'\Lambda_0 X_j + Z_{ij}'\beta_0 - U_{ij} \geq 0\right), \tag{58}$$

for $i \neq j$, $i, j = 1, \ldots, N$ and U_{ij} logistic.

According to democratic peace theory, democracies are less likely to engage in conflict with other democracies such that $\beta_0 < 0$ (e.g. Oneal and Russett, 1999). Note that any level (or monadic) effect of democracy on the propensity to initiate conflict generally is absorbed into the ego effects $\{A_i\}_{i=1}^N$ (out-degree effects), while any level effect on the propensity to be militarily targeted by others is absorbed into the alter effects $\{B_j\}_{j=1}^N$ (in-degree effects). Systematic cross-regional patterns in the costs and benefits of conflict are controlled for by the "homophily" term $X_i'\Lambda_0 X_j = W_{ij}'\lambda_0$.

The conditional likelihood of the network in hand, $\mathbf{D}$, here the observed pattern of conflict among nations, is

$$\begin{aligned} L\left(\mathbf{D};\delta_0,\beta_0\right) =& \prod_{i\neq j}\left[\frac{\exp\left(W_{ij}'\lambda_0 + R_i'\mathbf{A} + R_j'\mathbf{B} + Z_{ij}'\beta_0\right)}{1+\exp\left(W_{ij}'\lambda_0 + R_i'\mathbf{A} + R_j'\mathbf{B} + Z_{ij}'\beta_0\right)}\right]^{D_{ij}} \\ &\times\left[\frac{1}{1+\exp\left(W_{ij}'\lambda_0 + R_i'\mathbf{A} + R_j'\mathbf{B} + Z_{ij}'\beta_0\right)}\right]^{1-D_{ij}} \\ =& c\left(\mathbf{X},\mathbf{Z};\delta_0,\beta_0\right)\prod_{i\neq j}\exp\left(\mathbf{T}'\delta_0\right)\exp\left(\left[\sum_{i\neq j} D_{ij}Z_{ij}\right]'\beta_0\right) \end{aligned}$$

with R_i, as earlier, a $N\times 1$ vector with a 1 in its i^{th} element and zeros elsewhere and $c\left(\mathbf{X},\mathbf{Z};\delta_0,\beta_0\right)$ not varying with $\mathbf{D}$. Here $\mathbf{T}$ includes the vectorized cross-link matrix as well as the out- and in-degree sequences as discussed in the main text.

Let $\tau\left(\mathbf{v}\right)$ be the sufficient statistics for δ in network $\mathbf{v}$. Conditioning on $\mathbf{T}=\mathbf{t}$ yields

$$\begin{aligned} \Pr\left(\mathbf{D}=\mathbf{d}\middle|\,\mathbf{X}=\mathbf{x},\mathbf{Z}=\mathbf{z},\mathbf{T}=\mathbf{t};\delta_0,\beta_0\right) =& \frac{\prod_{i\neq j}\exp\left(\mathbf{t}'\delta_0\right)\exp\left(\left[\sum_{i\neq j} d_{ij}z_{ij}\right]'\beta_0\right)}{\sum_{\mathbf{v}\in\mathbb{D}_{\mathbf{s},\mathbf{m}}}\prod_{i\neq j}\exp\left(\tau\left(\mathbf{v}\right)'\delta_0\right)\exp\left(\left[\sum_{i\neq j} v_{ij}z_{ij}\right]'\beta_0\right)} \\ =& \frac{\exp\left(\left[\sum_{i\neq j} d_{ij}z_{ij}\right]'\beta_0\right)}{\sum_{\mathbf{v}\in\mathbb{D}_{\mathbf{s},\mathbf{m}}}\exp\left(\left[\sum_{i\neq j} v_{ij}z_{ij}\right]'\beta_0\right)} \\ =& \Pr\left(\mathbf{D}=\mathbf{d}\middle|\,\mathbf{Z}=\mathbf{z},\mathbf{T}=\mathbf{t},\beta_0\right), \end{aligned}$$

where the second equality follows from the fact that $\tau\left(\mathbf{v}\right)=\mathbf{t}$ for all $\mathbf{v}$ in $\mathbb{D}_{\mathbf{s},\mathbf{m}}$.

Taking logs yields, after some manipulation,

$$\begin{aligned} \ln\Pr\left(\mathbf{D}=\mathbf{d}\middle|\,\mathbf{Z}=\mathbf{z},\mathbf{T}=\mathbf{t},\beta_0\right) =& \left[\sum_{i\neq j} d_{ij}z_{ij}\right]'\beta_0 \\ &-\ln\left(\mathbb{E}\left[\exp\left(\left[\sum_{i\neq j} D_{ij}z_{ij}\right]'\beta_0\right)\right]\right) - \ln\left|\mathbb{D}_{\mathbf{s},\mathbf{m}}\right|, \end{aligned} \tag{59}$$

where the expectation in the second term to the right of the equality is taken with respect to the discrete uniform distribution on $\mathbb{D}_{\mathbf{s},\mathbf{m}}$. This expectation can be estimated using our simulation algorithm. The third term is invariant to β_0 and can consequently be ignored. Basing estimation and inference upon (59) allows a researcher to learn about β_0 in the presence of out- and in-degree heterogeneity as well as potentially complex patterns of homophily.

In our democratic peace theory example, Z_{ij} is binary. In this case $\sum_{i \neq j} D_{ij} Z_{ij}$ is simply a count of how many wars are initiated by democracies against other democracies. To compute the expectation in (59) we need an estimate of the distribution of this count induced by the discrete uniform distribution on $\mathbb{D}_{\mathbf{s},\mathbf{m}}$.